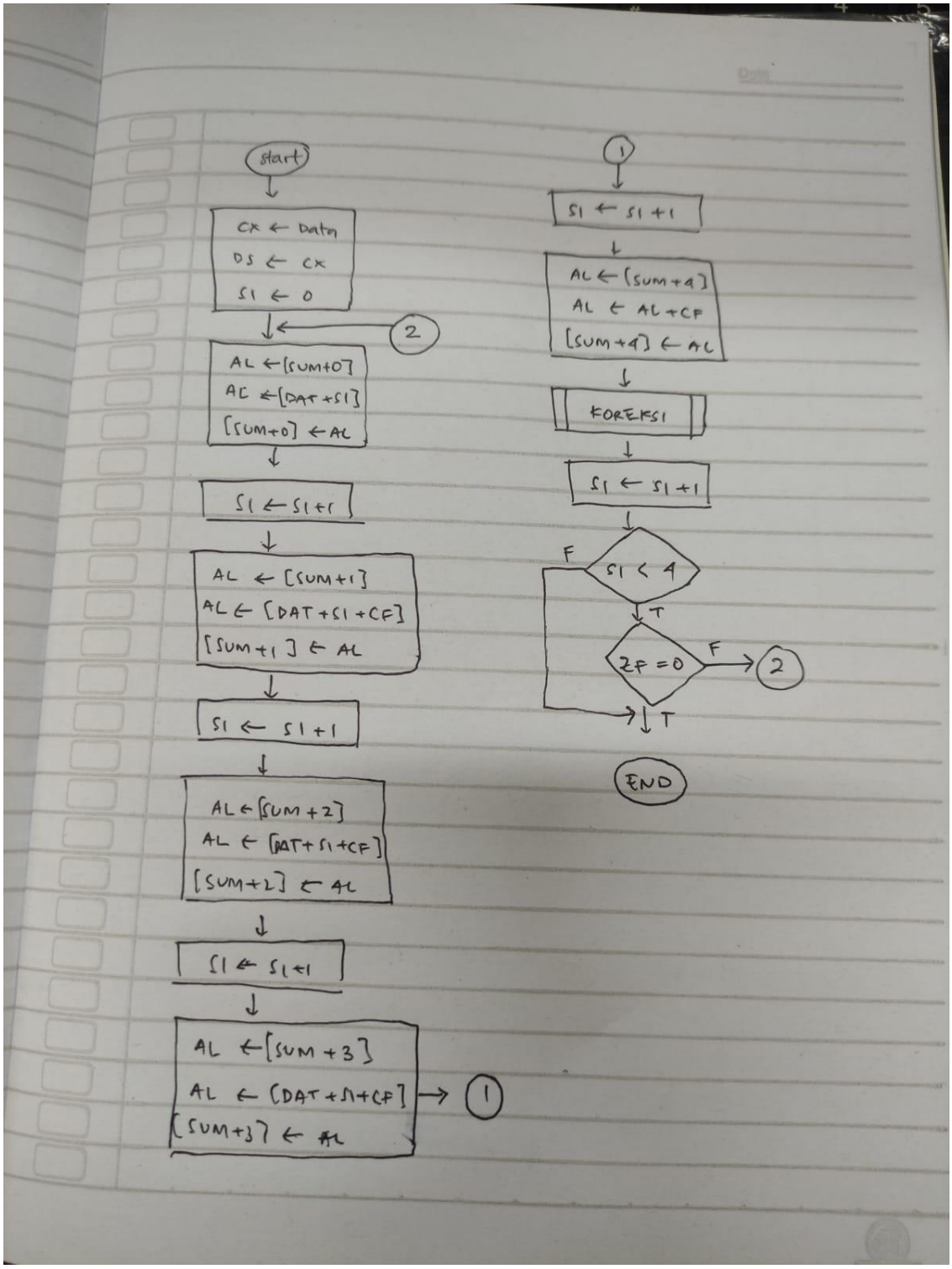
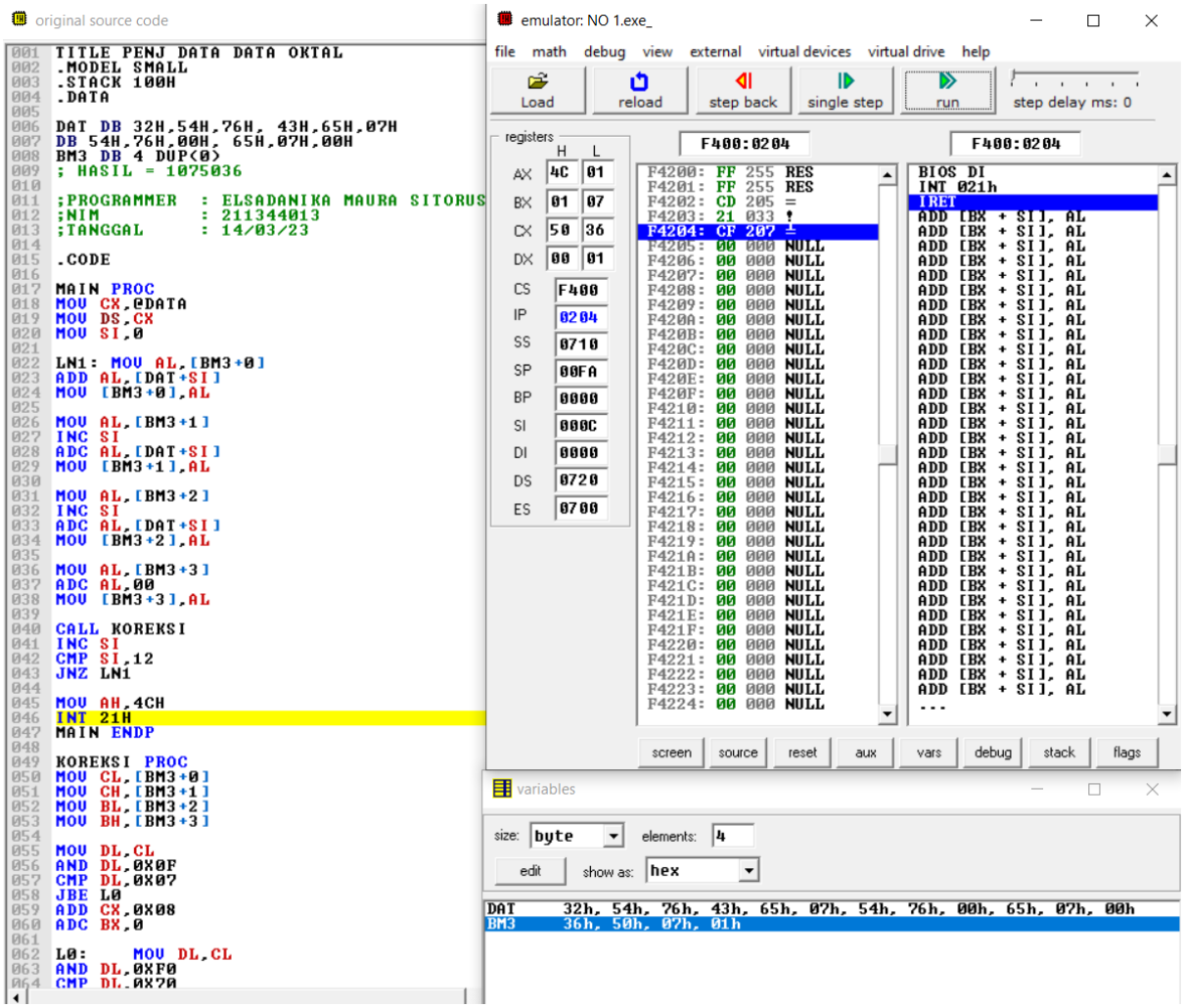
Nama : Elsadanika Maura Sitorus

NIM : 211344013 – 2NK

UTS LAB MIKRO

1) **Buat diagram alir dan program untuk menambahkan data-data oktal berikut: 765432 + 76543 + 7654 + 765.**






```
TITLE PENJ DATA DATA OKTAL
.MODEL SMALL
.STACK 100H
.DATA

DAT DB 32H,54H,76H, 43H,65H,07H
    DB 54H,76H,00H, 65H,07H,00H
BM3 DB 4 DUP(0)
; HASIL = 1075036

;PROGRAMMER   : ELSADANIKA MAURA SITORUS
;NIM          : 211344013
;TANGGAL      : 14/03/23

.CODE

MAIN PROC
    MOV CX,@DATA
    MOV DS,CX
    MOV SI,0

LN1: MOV AL,[BM3+0]
     ADD AL,[DAT+SI]
     MOV [BM3+0],AL

     MOV AL,[BM3+1]
     INC SI
     ADC AL,[DAT+SI]
     MOV [BM3+1],AL
```

```
     MOV AL,[BM3+2]
     INC SI
     ADC AL,[DAT+SI]
     MOV [BM3+2],AL

     MOV AL,[BM3+3]
     ADC AL,00
     MOV [BM3+3],AL

     CALL KOREKSI
     INC SI
     CMP SI,12
     JNZ LN1

    MOV AH,4CH
    INT 21H
    MAIN ENDP

KOREKSI PROC
        MOV CL,[BM3+0]
        MOV CH,[BM3+1]
        MOV BL,[BM3+2]
        MOV BH,[BM3+3]

        MOV DL,CL
        AND DL,0X0F
        CMP DL,0X07
        JBE L0
        ADD CX,0X08
        ADC BX,0

L0:    MOV DL,CL
        AND DL,0XF0
        CMP DL,0X70
        JBE L1
        ADD CX,0X80
        ADC BX,0

L1:     MOV DH,CH
        AND DH,0X0F
        CMP DH,0X07
        JBE L2
        ADD CH,0X08
        ADC BX,0

L2:     MOV DH,CH
        AND DH,0XF0
        CMP DH,0X70
        JBE L3
        ADD CH,0X80
        ADC BX,0

L3:     MOV DL,BL
        AND DL,0X0F
        CMP DL,0X07
        JBE L4
        ADD BL,0X08
        ADC BH,0

L4:     MOV DL,BL
        AND DL,0XF0
        CMP DL,0X70
        JBE L5
        ADD BL,0X80
        ADC BH,0

L5:     MOV DH,BH
```

```
        AND DH,0X0F
        CMP DH,0X07
        JBE L6
        ADD BH,0X08
        ADC AL,0

L6:     MOV DH,BH
        AND DH,0XF0
        CMP DH,0X70
        JBE L7
        ADD BH,0X80
        ADC AL,0

L7:     MOV DL,AL
        AND DL,0X0F
        CMP DL,0X07
        JBE L8
        ADD AL,0X08

L8:     MOV [BM3+0],CL
        MOV [BM3+1],CH
        MOV [BM3+2],BL
        MOV [BM3+3],BH

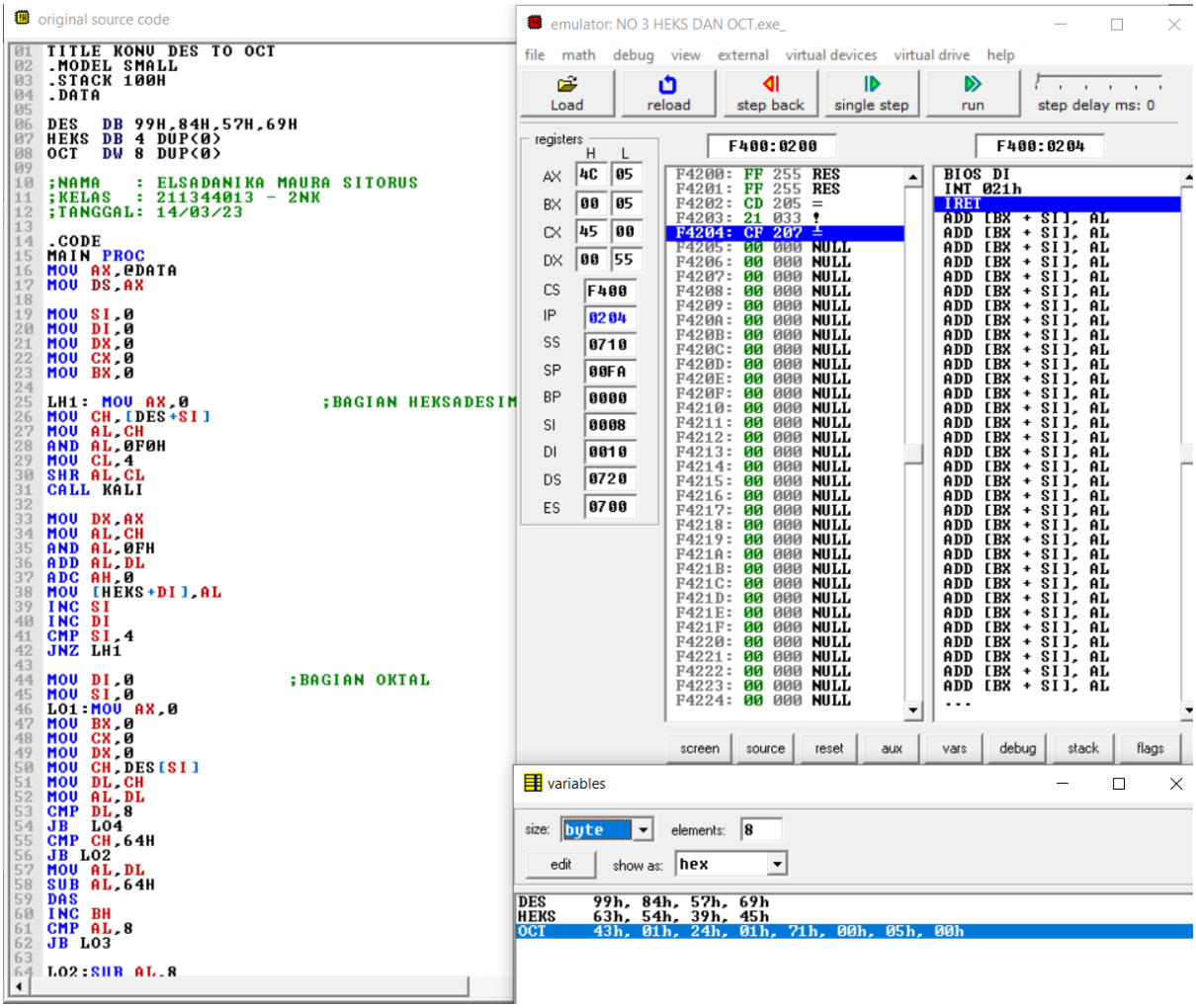
    RET
    KOREKSI ENDP

END MAIN
```

2) Buat program untuk menghitung banyaknya data bernilai negative, nol, positive, ganjil, dan genap dari data bilangan bertanda 8 bit berikut: AC, CA, A7, 63, 36, F3, 3F, D4, 4D, 9A, 80, 7F, 11, AA, 43, 8F, F8

-

3) Buat program untuk mengkonversi data decimal (99,84,57,69) menjadi heksadesimal dan oktal.




```
TITLE KONV DES TO OCT
.MODEL SMALL
.STACK 100H
.DATA

DES  DB 99H,84H,57H,69H
HEKS DB 4 DUP(0)
OCT  DW 8 DUP(0)

;NAMA    : ELSADANIKA MAURA SITORUS
;KELAS   : 211344013 - 2NK
;TANGGAL: 14/03/23

.CODE
MAIN PROC
    MOV AX,@DATA
    MOV DS,AX

    MOV SI,0
    MOV DI,0
    MOV DX,0
    MOV CX,0
    MOV BX,0

LH1: MOV AX,0              ;BAGIAN HEKSADESIMAL
    MOV CH,[DES+SI]
    MOV AL,CH
    AND AL,0F0H
    MOV CL,4
```

```
    SHR AL,CL
    CALL KALI

    MOV DX,AX
    MOV AL,CH
    AND AL,0FH
    ADD AL,DL
    ADC AH,0
    MOV [HEKS+DI],AL
    INC SI
    INC DI
    CMP SI,4
    JNZ LH1

    MOV DI,0              ;BAGIAN OKTAL
    MOV SI,0
LO1:MOV AX,0
    MOV BX,0
    MOV CX,0
    MOV DX,0
    MOV CH,DES[SI]
    MOV DL,CH
    MOV AL,DL
    CMP DL,8
    JB  LO4
    CMP CH,64H
    JB LO2
    MOV AL,DL
    SUB AL,64H
    DAS
    INC BH
    CMP AL,8
    JB LO3

LO2:SUB AL,8
    DAS
    INC BL
    CMP AL,8
    JAE LO2
    MOV DH,BH

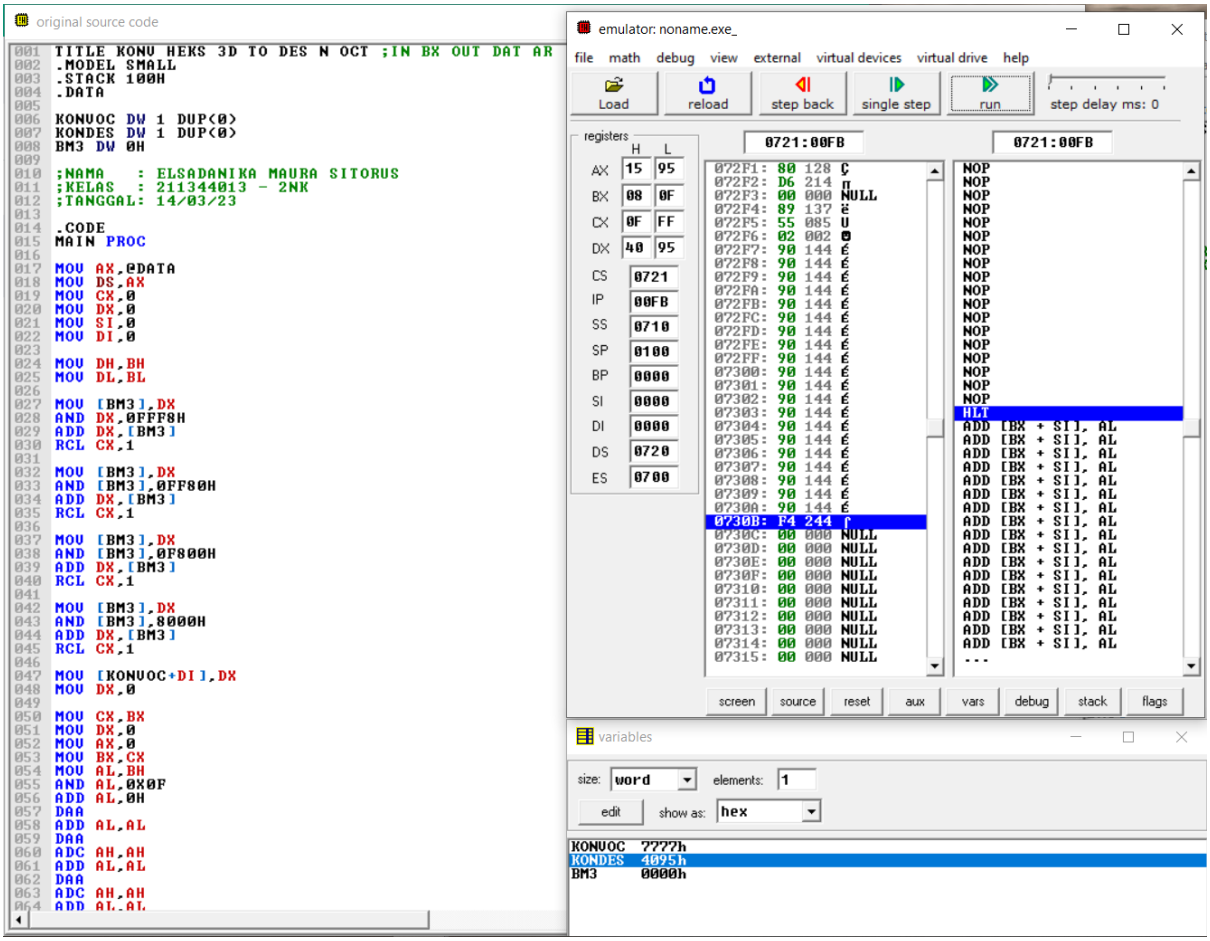
LO3:MOV DL,BL
    SHL DL,4
    ADD DL,AL
    MOV OCT[DI],DX
    ADD DI,2
    INC SI
    CMP SI,8
    JNZ LO1

LO4:MOV AH,4CH
    INT 21H
    MAIN ENDP

KALI PROC
    ADD AL,AL
    MOV BL,AL
    ADD AL,AL
    ADD AL,AL
    MOV BH,AL
    ADD BH,BL
    MOV AL,BH
    RET
    KALI ENDP

END MAIN
```

4) Buat program untuk mengkonversi data heksadesimal 3 digit yang ada pada register BX menjadi decimal dan oktal. Simpan hasil konversi pada data area.




```
TITLE KONV HEKS 3D TO DES N OCT ;IN BX OUT DAT AR
.MODEL SMALL
.STACK 100H
.DATA

KONVOC DW 1 DUP(0)
KONDES DW 1 DUP(0)
BM3 DW 0H

;NAMA    : ELSADANIKA MAURA SITORUS
;KELAS   : 211344013 - 2NK
;TANGGAL: 14/03/23

.CODE
MAIN PROC

     MOV AX,@DATA
     MOV DS,AX
     MOV CX,0
     MOV DX,0
     MOV SI,0
     MOV DI,0

     MOV DH,BH
     MOV DL,BL

     MOV [BM3],DX
     AND DX,0FFF8H
     ADD DX,[BM3]
```

```
 RCL CX,1

 MOV [BM3],DX
 AND [BM3],0FF80H
 ADD DX,[BM3]
 RCL CX,1

 MOV [BM3],DX
 AND [BM3],0F800H
 ADD DX,[BM3]
 RCL CX,1

 MOV [BM3],DX
 AND [BM3],8000H
 ADD DX,[BM3]
 RCL CX,1

 MOV [KONVOC+DI],DX
 MOV DX,0

MOV CX,BX
MOV DX,0
MOV AX,0
MOV BX,CX
MOV AL,BH
AND AL,0X0F
ADD AL,0H
DAA
ADD AL,AL
DAA
ADC AH,AH
ADD AL,AL
DAA
ADC AH,AH
ADD AL,AL
DAA
ADC AH,AH
ADD AL,AL
DAA
ADC AH,AH
MOV DX,AX;DL=256AN

AND BL,0XF0
SHR BL,1
SHR BL,1
SHR BL,1
SHR BL,1
MOV AL,BL
ADD AL,0H
DAA
MOV AH,AL;AH=16AN
MOV BH,0
ADD DL,AH
MOV AL,DL
ADD AL,0H
DAA
ADD AL,AL
DAA
ADC BH,BH
ADD AL,AL
DAA
ADC BH,BH
ADD AL,AL
DAA
ADC BH,BH
ADD AL,AL
DAA
ADC BH,BH
```

```
    MOV DL,AL
    MOV AL,BH
    ADD AL,0H
    DAA
    MOV BH,AL

    MOV AL,DH
    ADD AL,0H
    DAA
    ADC AL,AL
    DAA
    ADC AL,AL
    DAA
    ADC AL,AL
    DAA
    ADC AL,AL
    DAA
    MOV DH,AL
    ADD DH,BH
    MOV AL,DH
    ADD AL,0H
    DAA
    MOV DH,AL

    MOV AL,CL
    AND AL,0X0F
    ADD AL,0H
    DAA
    ADD DL,AL
    MOV AL,DL
    ADD AL,0H
    DAA
    MOV DL,AL
    ADC DH,0
    MOV [KONDES+DI],DX


END MAIN
```